

Module 05 Lab - Data Preparation **Objective:** To learn and apply the most common data preparation techniques. Raw data is rarely ready for a machine learning model.

- ✓ This process, also called preprocessing, is one of the most critical steps in the entire ML workflow. **In this lab, you will write more of the code.** Read the explanations and then complete the tasks in the code cells.

- ✓ Part 1: Setup and Initial Look We will continue using the Titanic dataset because it has the exact problems we need to solve: missing values and non-numeric data.

```
import pandas as pd
import numpy as np
# Load the dataset
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
# Let's look at the missing values
print("--- Missing Values Before Cleaning ---")
print(df.isnull().sum())
```

```
--- Missing Values Before Cleaning ---
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

Part 2: Handling Missing Values (Imputation) **Concept:** Most machine learning models cannot handle missing values (**NaN**). We must deal with them. Dropping the rows is an option, but you lose data. A better way is **imputation**, which means filling in the missing values with a calculated guess. Common imputation strategies:

- * **Mean:** Fill with the average value. Good for normally distributed data.
- * **Median:** Fill with the middle value. Better for skewed data or data with outliers (like **Fare**).
- * **Mode:** Fill with the most frequent value. Used for categorical data.

- ✓ Task 1: Impute the 'Age' Column The 'Age' column is missing many values. Since age can be skewed (e.g., by a few very old passengers), using the **median** is a robust choice. **Your Task:** Calculate the median of the 'Age' column and use the `.fillna()` method to replace the missing values.

```
# --- ENTER YOUR CODE HERE ---

# 1. Calculate the median of the 'Age' column
median_age = df['Age'].median()

# 2. Fill the missing values in 'Age' with the median
df['Age'].fillna(median_age, inplace=True)

# 3. Verify that there are no more missing values in 'Age'
print("Missing values in 'Age' after imputation:")
print(df['Age'].isnull().sum())

Missing values in 'Age' after imputation:
0
```

```

/tmp/ipykernel_1233/1039361305.py:7: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d

df['Age'].fillna(median_age, inplace=True)

```

Part 3: Encoding Categorical Features
Concept: Machine learning models are mathematical, so they need numbers, not text. We need to convert categorical columns (like 'Sex' and 'Embarked') into a numerical format. The most common method is **One-Hot Encoding**. One-Hot Encoding takes a column with **N** categories and turns it into **N** new columns, each with a **1** or **0**. For example, the 'Sex' column (**male**, **female**) becomes two new columns: **Sex_male** and **Sex_female**. Pandas has a convenient function called `pd.get_dummies()` that does this for us.

Task 2: One-Hot Encode Categorical Columns
Your Task: Use `pd.get_dummies()` to encode the 'Sex' and 'Embarked' columns. Make sure to drop the original columns after encoding.

```

# --- ENTER YOUR CODE HERE ---

# 1. Calculate the median of the 'Age' column
median_age = df['Age'].median()

# 2. Fill the missing values in 'Age' with the median
df['Age'].fillna(median_age, inplace=True)

# 3. Verify that there are no more missing values in 'Age'
print("Missing values in 'Age' after imputation:")
print(df['Age'].isnull().sum())

Missing values in 'Age' after imputation:
0
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```

Double-click (or enter) to edit

Part 4: Feature Scaling
Concept: Many models are sensitive to the scale of the features. For example, **Age** (from 0-80) and **Fare** (from 0-512) are on very different scales. This can cause the model to incorrectly believe that **Fare** is a more important feature simply because its values are larger. **Feature Scaling** solves this by putting all features on a similar scale. A common method is **Standardization** (`StandardScaler` in scikit-learn), which rescales the data to have a mean of 0 and a standard deviation of 1. **Important:** You only scale your numerical features, not your target variable or your newly encoded categorical columns.

Task 3: Scale the 'Age' and 'Fare' Columns
Your Task: Use `StandardScaler` from `sklearn.preprocessing` to scale the 'Age' and 'Fare' columns.

```

# --- ENTER YOUR CODE HERE ---

```

```
# 1. Use get_dummies to create new columns for 'Sex' and 'Embarked'
# Set `drop_first=True` to avoid multicollinearity (a statistical issue), which drops one of the new columns (e.g
encoded_df = pd.get_dummies(df, columns=['Sex', 'Embarked'], drop_first=True)

# 2. Display the first few rows of the new DataFrame to see the new columns
print(encoded_df.head())
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Age	SibSp	Parch	\
0	Braund, Mr. Owen Harris	22.0	1	0	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0	
2	Heikkinen, Miss. Laina	26.0	0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0	
4	Allen, Mr. William Henry	35.0	0	0	

	Ticket	Fare	Cabin	Sex_male	Embarked_Q	Embarked_S
0	A/5 21171	7.2500	NaN	True	False	True
1	PC 17599	71.2833	C85	False	False	False
2	STON/O2. 3101282	7.9250	NaN	False	False	True
3	113803	53.1000	C123	False	False	True
4	373450	8.0500	NaN	True	False	True

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```
## 📌 Knowledge Check **Instructions:** Answer the following
questions in this markdown cell. **1.Why is it often better to
impute missing values with the median instead of the mean?**
The mean is quite sensitive to outliers. In case there are
some very large numbers in a dataset (such as some very old
ages), then the mean will be skewed towards those numbers. On
the other hand, the median is the number that lies exactly at
the center of the sorted data set, and therefore, it is not
affected by outliers. **2.Explain in your own words what
One-Hot Encoding does and why it is necessary.** One-Hot
encoding is a process where categorical data (such as text
data) is converted into numerical data using a separate column
of binary (0s and 1s) for each individual category. This step
is essential due to the following two reasons: Machine
learning models need numerical data in order to solve the
equation mathematically.This step makes sure that the machine
learning model does not consider a mathematical order/ranking
among different categories. For instance, assigning Cherbourg
= 1, Queenstown = 2, and Southampton = 3 would make the
machine learning model think Southampton is mathematically
greater than Cherbourg. **3. Would you need to apply Feature
Scaling to a Decision Tree model? Why or why not? (Hint: Think
about how a Decision Tree makes its splits).** No, feature
scaling is not needed for Decision Trees because Decision
Trees split the data by using simple logical splits on a
threshold basis for each feature one after another, like Age
greater than 30 or Fare less than or equal to 50. Since
decision trees evaluate the features independently to get the
best split point, the actual number values or magnitudes do
not matter.
```

📌 Knowledge Check **Instructions:** Answer the following questions in this markdown cell.

1.Why is it often better to impute missing values with the median instead of the mean? The mean is quite sensitive to outliers. In case there are some very large numbers in a dataset (such as some very old ages), then the mean will be skewed towards those numbers. On the other hand, the median is the number that lies exactly at the center of the sorted data set, and therefore, it is not affected by outliers.

2.Explain in your own words what One-Hot Encoding does and why it is necessary. One-Hot encoding is a process where categorical data (such as text data) is converted into numerical data using a separate column of binary (0s and 1s) for each individual category. This step is essential due to the following two reasons: Machine learning models need numerical data in order to solve the equation mathematically.This step makes sure that the machine learning model does not consider a mathematical order/ranking among different categories. For instance, assigning Cherbourg = 1, Queenstown = 2, and Southampton = 3

would make the machine learning model think Southampton is mathematically greater than Cherbourg. **3. Would you need to apply Feature Scaling to a Decision Tree model? Why or why not? (Hint: Think about how a Decision Tree makes its splits).** No, feature scaling is not needed for Decision Trees because Decision Trees split the data by using simple logical splits on a threshold basis for each feature one after another, like Age greater than 30 or Fare less than or equal to 50. Since decision trees evaluate the features independently to get the best split point, the actual number values or magnitudes do not matter.